1.0 INTRODUCTION TO GIS

1.1 Preamble

A **geographic information system (GIS)** is a computer-based system designed to capture, store, manipulate, analyze, manage, and present all types of spatial or geographical data.

GIS is a set of tools (or technology) that allow for the processing of ***spatial data*** into ***information***. This set of tools is open ended, but will include data input, data storage, data manipulation, and a reporting system.

The GIS is sometimes used for **geographical information science** or **geospatial information studies** to refer to the academic discipline or career of working with geographic **information systems** and is a large domain within the broader academic discipline of **Geoinformatics**.

GIS lets us visualize, question, analyze, and interpret data to understand relationships, patterns, and trends. GIS benefits organizations of all sizes and in almost every industry. There is a growing interest in and awareness of the economic and strategic value of GIS

In a general sense, the term describes any **information system** that integrates, stores, edits, analyzes, shares, and displays **geographic** information. Information Systems allows the transformation of data into information via: structuring, formatting, conversion and modeling

GIS: transforms data with a spatial component. A spatial components here refer to location- based data, with coordinates (E,N,H or Latitude and longitude), on, below or above earths surface.

The data with the "where".

GIS technology has been called by different names like: Geospatial Information Systems, Computer Aided Mapping, Automated Cartography, Land Information Systems, Environmental Information Systems, etc

1.2 Definition of GIS

EXAM

Geographical Information System has been defined differently by different authors

§ "…a system of hardware, software, and procedures designed to support the capture, management, manipulation, analysis, modeling, and display of spatially referenced data for solving complex planning & management problems" (Rhind, 1989)

§ "…a computer system capable of assembling, storing, manipulating, and displaying geographically referenced information…" (USGS, 1997)

§ "…a set of computer-based systems for managing geographic data and using those data to solve spatial problems" (Lo & Yeung, 2002)

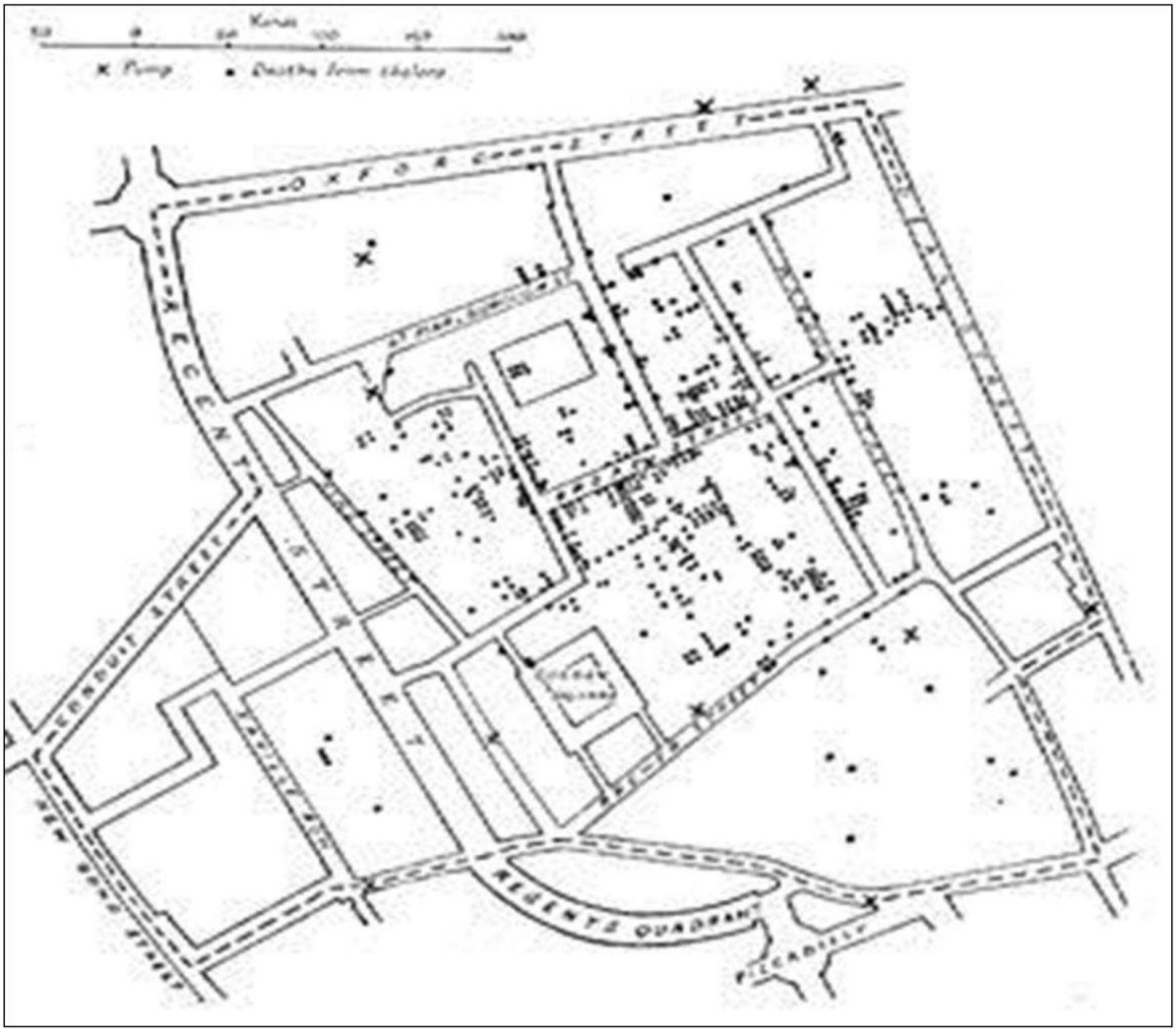
§ a computer system that allows the analysis and display of data with a spatial component (Phillips, 2002)

In general it can be defined as a ***"computer-based system that support capture, management, manipulation, analysis, modeling, and display of***

spatially referenced data that provides planning, management, development and humanitarian solutions”.

1.3 History of Geographic Information System

The first known use of the term "**geographic information system**" was by **Roger Tomlinson** in the year 1968 in his paper "A Geographic Information System for Regional Planning". Tomlinson is also acknowledged as the "**father of GIS**".



***E. W. Gilbert's** version (1958) of **John Snow**'s 1855 map of the Soho cholera outbreak showing the clusters of cholera cases in the **London** epidemic of 1854*

Previously, one of the first applications of spatial analysis in **epidemiology** is the 1832. The **French geographer Charles Picquet** represented the 48 districts of the city of Paris by halftone color gradient according to the percentage of deaths by cholera per 1,000 inhabitants.

In 1854 **John Snow** depicted a **cholera** outbreak in **London** using points to represent the locations of some individual cases, an early successful use of a geographic methodology in epidemiology. While the basic elements of **topography** and theme existed previously in **cartography**, the John Snow map was unique, using cartographic methods not only to depict but also to analyze clusters of geographically dependent phenomena.

The early 20th century saw the development of **photozincography**, which allowed maps to be split into layers, for example one layer for vegetation and another for water. This was particularly used for printing contours – drawing these was a labor-intensive task but having them on a separate layer meant they could be worked on without the other layers to confuse the

draughtsman. This work was originally drawn on glass plates but later **plastic film** was introduced, with the advantages of being lighter, using less storage space and being less brittle, among others. When all the layers were finished, they were combined into one image using a large process camera. Once color printing came in, the layers idea was also used for creating separate printing plates for each color. While the use of layers much later became one of the main typical features of a contemporary GIS, the photographic process just described is not considered to be a GIS in itself – as the maps were just images with no database to link them to.

Computer hardware development spurred by **nuclear weapon** research led to general purpose computer "mapping" applications by the early

1960s.The year 1960 saw the development of the world's first true operational GIS in **Ottawa, Ontario**, Canada by the federal Department of Forestry and Rural Development. Developed by **Dr. Roger Tomlinson**, it was called the **Canada Geographic Information System** (CGIS) and was used to store, analyze, and manipulate data collected for the Canada Land Inventory – an effort to determine the land capability for rural Canada by mapping information about **soils, agriculture, recreation, wildlife, waterfowl, forestry and land use** at a scale of **1:50,000**. A rating classification factor was also added to permit analysis.

CGIS was an improvement over "computer mapping" applications as it provided capabilities for **overlay, measurement, and digitizing/scanning**. It supported a national coordinate system that spanned the continent, coded lines as **arcs** having a true embedded **topology** and it stored the attribute and locational information in separate files. As a result of this, Tomlinson has become known as the "father of GIS", particularly for his use of overlays in promoting the spatial analysis of convergent geographic data.

CGIS lasted into the 1990s and built a large digital land resource database in Canada. It was developed as a **mainframe**-based system in support of federal and provincial resource planning and management. Its strength was continent-wide analysis of complex **datasets**. The CGIS was never available commercially.

In 1964 Howard T. Fisher formed the Laboratory for Computer Graphics and Spatial Analysis at the **Harvard Graduate School of Design** (LCGSA 1965–1991), where a number of important theoretical concepts in spatial data handling were developed, and which by the 1970s had distributed seminal software code and systems, such as SYMAP, GRID, and

ODYSSEY that served as sources for subsequent commercial development to universities, research centers and corporations worldwide.

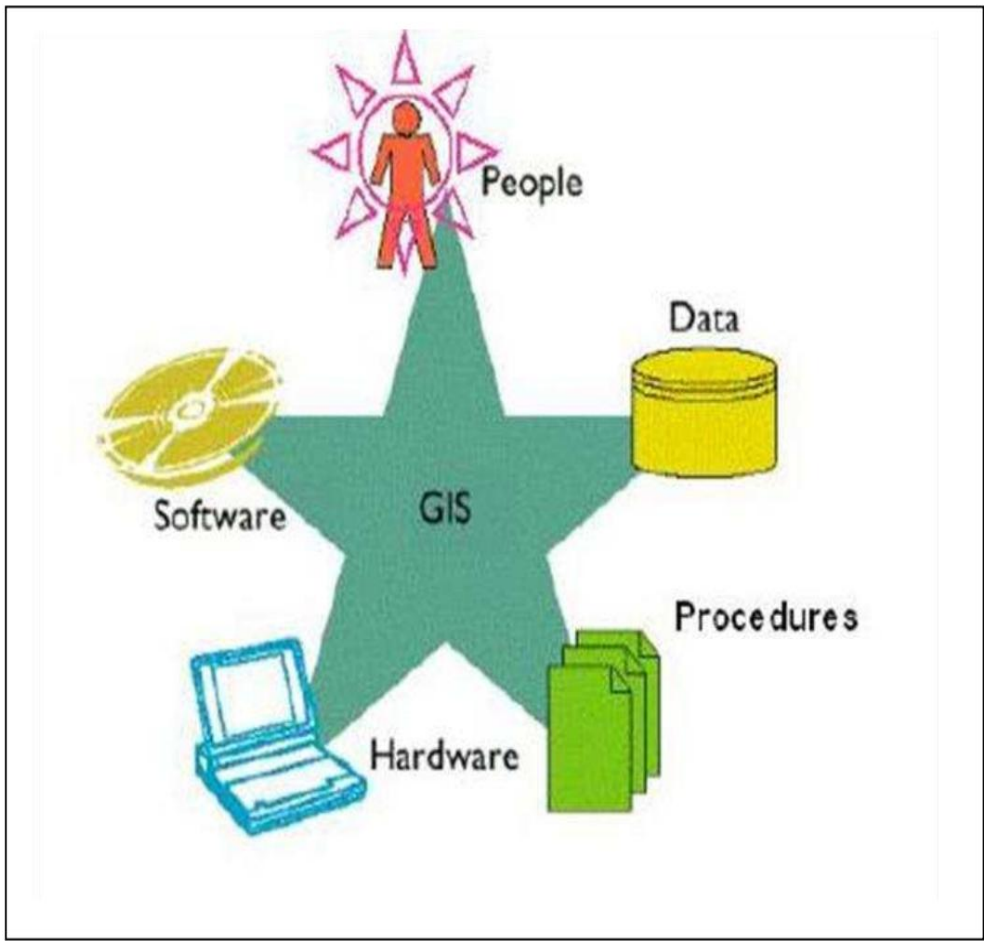
By the early 1980s, **M&S** Computing (later **Intergraph**) along with Bentley Systems Incorporated for the **CAD** platform, Environmental Systems Research Institute **(ESRI**), **CARIS** (Computer Aided Resource Information System), **MapInfo Corporation** and **ERDAS** (Earth Resource Data Analysis System) emerged as **commercial vendors** of GIS software, successfully incorporating many of the CGIS features, combining the **first generation approach to separation of spatial and attribute information with a second generation approach to organizing attribute data into database structures**. In parallel, the development of two public domain systems (**MOSS** and **GRASS GIS**) began in the late 1970s and early 1980s.

In 1986, Mapping Display and Analysis System (**MIDAS**), the first desktop GIS product emerged for the **DOS** operating system. This was renamed in 1990 to MapInfo for Windows when it was ported to the **Microsoft Windows** platform. This began the process of moving GIS from the research department into the business environment.

By the end of the 20th century, the rapid growth in various systems had been consolidated and standardized on relatively few platforms and users were beginning to explore viewing GIS data over the **Internet**, requiring data format and transfer standards. More recently, a growing number of **free, open-source GIS packages** run on a range of operating systems and can be customized to perform specific tasks. Increasingly **geospatial data** and **mapping applications** are being made available via the **world wide web.**

1.4 Components of GIS

An operational GIS also has a series of components that combine to make the system work. These components are critical to a successful GIS. There include: Hardware, software, data, people, methods/procedures



- **Hardware:** Hardware is the computer system on which a GIS operates. Today, GIS software runs on a wide range of hardware types, from centralized computer servers to desktop computers used in stand-alone or networked configurations.

- **Software :** GIS software provides the functions and tools needed to store, analyze, and display geographic information. A review of the key GIS software subsystems is provided above.

- **Data:** Perhaps the most important component of a GIS is the data. Geographic data and related tabular data can be collected in-house, compiled to custom specifications and requirements, or occasionally purchased from a commercial data provider. A GIS can integrate spatial data with other existing data resources, often stored in a corporate DBMS. The integration of spatial data (often proprietary to the GIS software), and tabular data stored in a DBMS is a key functionality afforded by GIS.

- **People:** GIS technology is of limited value without the people who manage the system and develop plans for applying it to real world problems. GIS users range from technical specialists who design and maintain the system to those who use it to help them perform their everyday work. The identification of GIS specialists versus end users is often critical to the proper implementation of GIS technology.

- **Methods:** A successful GIS operates according to a well-designed implementation plan and business rules, which are the models and operating practices unique to each organization.

1.5 GIS Subsystems and Functions

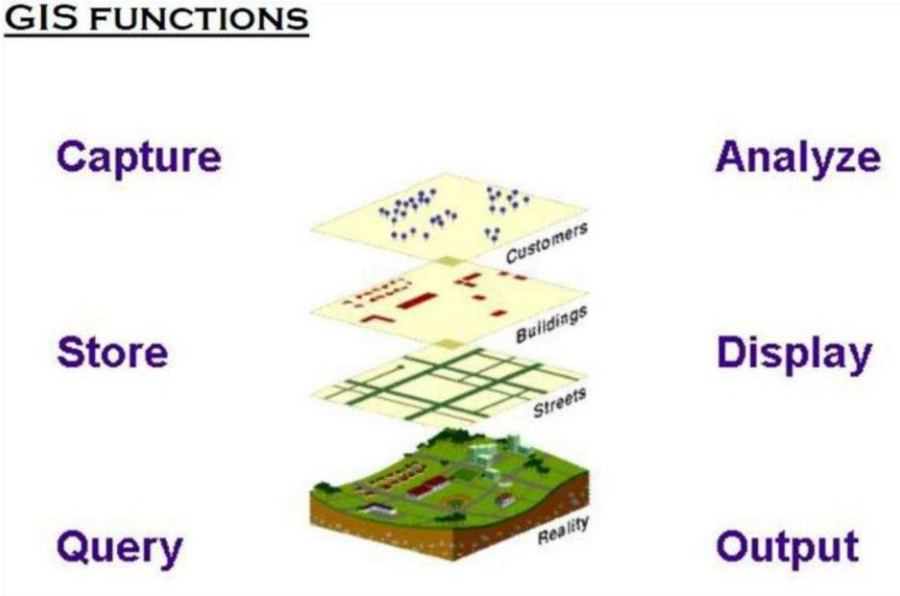
A GIS software has **four** main functional subsystems. These are:

→ a data input subsystem;

→ a data storage and retrieval subsystem;

→ a data manipulation and analysis subsystem; and

→ a data output and display subsystem.



- Data Input: A data input subsystem allows the user to capture, collect, and transform spatial and thematic data into digital form. The data inputs are usually derived from a combination of hard copy maps, aerial photographs, remotely sensed images, reports, survey documents, etc.

- Data Storage and Retrieval: The data storage and retrieval subsystem organizes the data, spatial and attribute, in a form which permits it to be quickly retrieved by the user for analysis, and

permits rapid and accurate updates to be made to the database. This component usually involves use of a database management system (DBMS) for maintaining attribute data. Spatial data is usually encoded and maintained in a proprietary file format.

- **Data Manipulation and Analysis:** The data manipulation and analysis subsystem allows the user to define and execute spatial and attribute procedures to generate derived information. This subsystem is commonly thought of as the *heart of a GIS*, and usually distinguishes it from other database information systems and computer-aided drafting (CAD) systems.

- **Data Output:** The data output subsystem allows the user to generate graphic displays, normally maps, and tabular reports representing derived information products.

The critical function for a GIS is, by design, the analysis of spatial data.

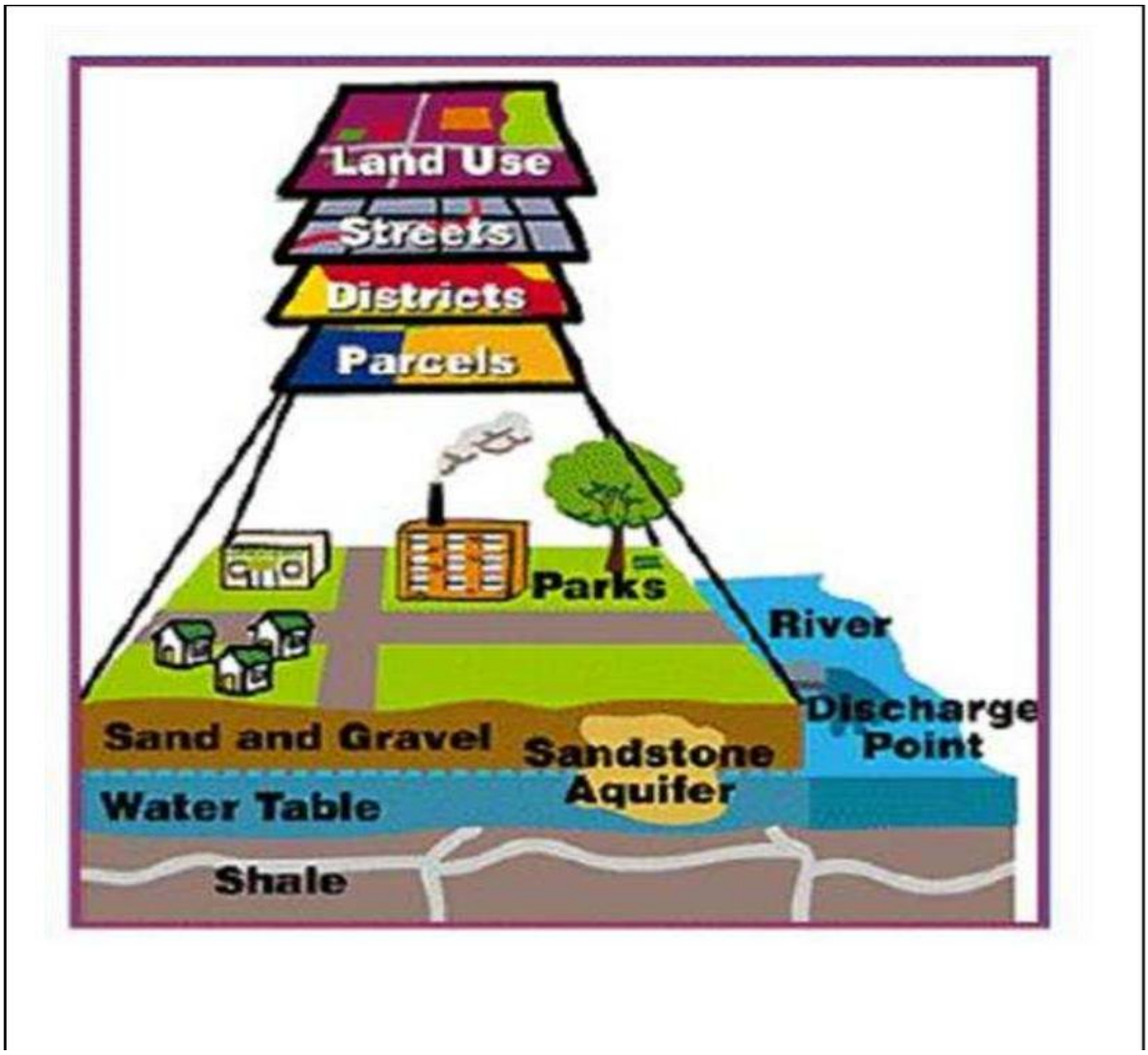
Basic GIS functions are contained in the definition and include among others: data capture, storage, query, analysis, display and output/presentation or visualization.

1.6 GIS Data Models (Organization)

A GIS stores information about the world as a collection of thematic layers that can be linked together by geography. This simple but extremely powerful and versatile concept has proven invaluable for solving many real-world problems from tracking delivery vehicles, to recording details of planning applications, to modeling global atmospheric circulation.

The thematic layer approach allows us to organize the complexity of the real world into a simple representation to help facilitate our understanding of natural relationships.

Layers represents specific feature class like: roads/streets, parcels of land, District boundary and land use



1.7 Some Advantages of GIS

1) Cost Savings from Greater Efficiency:

Typical implementations can result in a savings of 10 to 30 percent in operational expenses through reduction in fuel use and staff time, improved customer service, and more efficient scheduling. ***GIS helped the City of Woodland refine its fleet scheduling, saving fuel and labour.***

2) Better Decision Making:

GIS is the go-to technology for making better decisions about location. Common examples include real estate site selection, route/corridor selection, evacuation planning, conservation, natural resource extraction, etc. Making correct decisions about location is critical to the success o f an organization. For instance, ***GIS-based disaster decision support system helps Taiwan plan for and respond to***m ***typhoons***.

3) Improved Communication:

GIS-based maps and visualizations greatly assist in understanding situations and in storytelling. They are a type of language that improves communication between different teams, departments, disciplines, professional fields, organizations, and the public. ***Michels Corporation improved collaboration and communication with GIS***

4) Managing Geographically:

GIS is becoming essential to understanding what is happening and what will happen in geographic space. Once we understand, we can prescribe action. This new approach to management—managing geographically—is transforming the way organizations operate. ***Kuwait University used GIS to design and build a multibillion- dollar expansion***.

1.8 Application Areas

GIS are now used extensively in government, military, medical, industry, business, and academic/research for a wide range of applications including environmental resource analysis, land use planning, locational analysis, tax appraisal, utility and infrastructure planning, real estate analysis, marketing and demographic analysis, habitat studies, and archaeological analysis.

❖ One of the first major areas of application was in **natural resources management**, including management of

- wildlife habitat,
- wild and scenic rivers,
- recreation resources,
- floodplains,
- wetlands,
- agricultural lands,
- aquifers,
- forests.

❖ One of the largest areas of application has been in **facilities management**. Uses for GIS in this area have included

- locating underground pipes and cables,
- balancing loads in electrical networks,
- planning facility maintenance,
- tracking energy use.

❖ Local, state, and federal governments have found GIS particularly useful in **land management**. GIS has been commonly applied in areas like

- zoning and subdivision planning,
- land acquisition,
- environmental impact policy,
- water quality management,
- Maintenance of ownership.

❖ More recent and innovative uses of GIS have used information based on **street-networks**. GIS has been found to be particularly useful in:

- address matching,
- location analysis or site selection,
- development of evacuation plans.

1.9 GIS Users

§ Specialist: includes programmers, designers, developers

§ General Users: planners, medics, scientists, administrators, teachers (us)

§ Viewers: everyone (our "clients")

Examples of Applied GIS

- **Urban Planning, Management & Policy**
  - Zoning, subdivision planning
  - Land acquisition
  - Economic development
  - Code enforcement
  - Housing renovation programs
  - Emergency response
  - Crime analysis
  - Tax assessment
- **Environmental Sciences**
  - Monitoring environmental risk
  - Modeling stormwater runoff
  - Management of watersheds, floodplains, wetlands, forests, aquifers
  - Environmental Impact Analysis
  - Hazardous or toxic facility siting
  - Groundwater modeling and contamination tracking
- **Political Science**
  - Redistricting
  - Analysis of election results
  - Predictive modeling
- **Civil Engineering/Utility**
  - Locating underground facilities
  - Designing alignment for freeways, transit
  - Coordination of infrastructure maintenance
- **Business**
  - Demographic Analysis
  - Market Penetration/ Share Analysis
  - Site Selection
- **Education Administration**
  - Attendance Area Maintenance
  - Enrollment Projections
  - School Bus Routing
- **Real Estate**
  - Neighborhood land prices
  - Traffic Impact Analysis
  - Determination of Highest and Best Use
- **Health Care**
  - Epidemiology
  - Needs Analysis
  - Service Inventory